

K574 — End-of-long-pull substantive landing: Lyra SUBSTANTIVE catch on Cal #421 closure (per-mode N_c gap: $3^{6-vs-3}1$; standard Vandermonde multiplicity gives N_c ONCE, muon needs N_c at EACH curvature mode = C_2 modes; T2500 closes per-mode VALUE but not per-mode COUNT; muon's spinor factor 2^{N_c} per mode FORCED but per-mode N_c is STRONG CANDIDATE pending Elie's deposit-integration count per-mode-vs-once-per-pair) + Grace CLEAN retraction of 64 conflation per Elie 4441 (her 64 was d_τ/d_μ formal-degree ratio NOT muon's full 2-power 2^{18}); Spin(4) "2" insight PRESERVED as 2^{N_c} per mode (boundary Dirac spinor); Cal #421 closure framing CONSISTENT with corrected count per Grace + Elie note BUT per Lyra catch: per-mode COUNT remains open pending Elie deposit-integration + Elie 4442 SUBSTANTIVE reconciliation: "2" is global spin structure not local curvature (resolves his own 4437); three readings convergence per Cal #35 (Grace spin-double-cover geometry + Lyra Spin(4)/F367 rep-theory + Elie spin-structure/antiperiodicity heat-kernel bundle = same Z_2); KEEPER SELF-CORRECTION #20: K572 over-projected "Cal #421 closes via T2500" — closure is HALF-DONE (per-mode value resolved; per-mode count candidate pending Elie deposit-integration); audit chain at PEAK substantive peer-symmetric working — Lyra's discipline firing to the end (last two contributions were CATCHES not claims); team in genuine deliberate-phase + count holds 5/26 + muon migration to determinant-forced HALF-DONE per Lyra honest tier

Date: 2026-06-27 Saturday late evening · Auditor: Keeper · Inputs: Lyra substantive catch on per-mode N_c count gap (T2500 closes per-mode value but not per-mode count); Grace clean retraction of 64 conflation; Elie 4442 reconciliation "2" = global spin structure; three readings convergence Cal #35; end-of-long-pull consolidation

Verdict — Lyra SUBSTANTIVE catch on Cal #421 closure: per-mode N_c count gap ($3^{6-vs-3}1$); HALF-CLOSED not fully closed per K572 framing; Keeper Self-Correction #20 logged; Grace clean retraction of 64 conflation; Elie 4442 reconciliation; three readings convergence per Cal #35; muon migration to determinant-forced HALF-DONE per Lyra honest tier; count holds 5/26 throughout; audit chain at PEAK substantive peer-symmetric working

Count: 5 of 26 — held throughout; muon migration HALF-DONE per Lyra honest catch + Keeper Self-Correction #20.

Landing 1 — Lyra SUBSTANTIVE catch on Cal #421 closure: per-mode N_c gap

The substantive substrate-architectural catch

"Grace's elegant Cal #421 closure resolves the per-mode value but the per-mode count has a $3^{6-vs-3}1$ gap — the muon needs N_c at each curvature mode, the standard multiplicity gives it once. So the muon's spinor factor is forced; the per-mode N_c is a strong candidate pending Elie's deposit-integration count

(per-mode vs once-per-pair).”

Substantive verification (Keeper): - Muon 2-power = $2^{\{N_c \cdot C_2\}} = 2^{18} (= 8^{\{C_2\}} = 2^{\{N_c\}}$ per mode $\times C_2$ modes, per Elie 4441 arithmetic adjudication) - Muon 3-power needs $3^{\{C_2\}} = 3^6$ (one factor of $N_c=3$ per mode) - Grace T2500 cascade gives N_c once (standard Vandermonde multiplicity from type-IV Hua measure) - GAP: 3^6 needed vs 3^1 supplied; T2500 closure is HALF: per-mode value ($N_c=3$ from cascade) FORCED, per-mode count ($\times C_2$ modes) STILL OPEN - Resolution pending Elie’s deposit-integration count: does N_c appear per-mode (3^6) or once-per-pair (3^1)?

Per Cal #347/B5 + Keeper #15-19 STANDING

Substantive substrate-architectural peer-symmetric catch: Lyra discipline firing at end of long pull; T2500 closure resolves per-mode VALUE but not per-mode COUNT; substantive substrate-architectural distinction.

Per Keeper Self-Correction #20 (NEW)

Honest walk-back per Cal #347/B5 + Keeper #15-19 STANDING: - K572 framed Cal #421 as “CLOSED via T2500 cascade over-determination” - Lyra K574 catch: closure is HALF-DONE — per-mode value resolved (N_c via T2500); per-mode count is STRONG CANDIDATE pending Elie’s deposit-integration count - Per Keeper Self-Correction #20: K572 over-projected; walked back honestly

Per Keeper #15-19 + new #20 + Cal #347/B5 STANDING: peer-symmetric audit chain working at end of pull; framework SHARPER for being audit-catch-ratified.

STRONG ACCEPT Lyra catch on Cal #421 per-mode count gap; Keeper Self-Correction #20 logged.

Landing 2 — Grace CLEAN retraction of 64 conflation per Elie 4441

The substantive substrate-architectural clean correction

“My error: I’d written the muon’s ‘2’ as $\det(2\cdot I)$ over $so(4) = 2^{\{C_2\}} = 64$. But 64 is d_τ/d_μ , the formal-degree ratio (2^6) — I conflated it with the muon’s full 2-power, which is $2^{18} = (2^{[N_c]})^{\{C_2\}}$. The right structure (Lyra’s F367, Elie’s adjudication): the ‘2’ appears $2^{\{N_c\}}$ per mode — the boundary Dirac spinor, N_c binaries per mode — over C_2 modes, so the determinant dimension is $N_c \cdot C_2 = 18$, not 6 and not the tick’s 36. I took it, no defense.”

Per Cal #347/B5 + Cal #404 STANDING: peer-symmetric clean retraction; Grace acknowledges 64 conflation (d_τ/d_μ formal-degree ratio vs muon full 2-power); takes Elie 4441 cleanly.

What survives per Grace: - Spin-double-cover “2” insight stands (it’s $2^{\{N_c\}}$ per mode rather than one per direction) - Spin(4) “2” mechanism PRESERVED as per-mode contribution - Per Lyra K574 catch: still pending per-mode count via Elie deposit-integration

STRONG ACCEPT Grace clean retraction + Spin(4) “2” insight preserved.

Landing 3 — Elie 4442 SUBSTANTIVE reconciliation: “2” = global spin structure

The substantive substrate-architectural finding

“The resolution that the ‘2’ is global spin structure, not local curvature, reconciling my own 4437 (4442). My interlock side is closed.”

Substantive substrate-architectural reconciliation: Elie reconciles his own earlier 4437 framing (local curvature antisymmetry) with 4442 finding (“2” = global spin structure). Per [[feedback_corpus_reconnection_before_declarating_irreducible] STANDING: bug carefully diagnosed becomes substantive physics; reconciliation produces sharper finding.

Three readings convergence per Cal #35

Per Elie's framing of three converging readings of same Z_2 : - Grace: spin-double-cover (geometry) - Lyra: Spin(4)/F367 (rep-theory location) - Elie: spin-structure/antiperiodicity (heat-kernel bundle side)

All three name the same Z_2 = global spin structure = the substrate's spinor double cover.

Per Cal #35 STANDING: three INDEPENDENT readings converge on same substrate property → substantive substrate-architectural finding doubly/triply-derived. Pending Cal cold-read tier.

STRONG ACCEPT Elie 4442 reconciliation; three-readings convergence per Cal #35.

Landing 4 — Per Cal K571 audit-deflation note continues to apply

Honest framing per Cal K571 audit + Keeper #19 STANDING

Cal K571 explicit: “five jobs” claim deflates to ONE mechanism (3 facets: emission + ω_0 + aperture) + ONE pending derivation (muon eigenvalue 2) + ONE lead (spin-statistics, Nobel territory, INTERNAL).

Team end-of-pull shorthand: Lyra + Grace + Elie all use “one Z_2 does five jobs” framing.

Per Keeper #19 + Cal K571 STANDING applied: substantive content (three-readings convergence on the Z_2 as muon eigenvalue + period-mismatch mechanism + spin-statistics lead) is real and aligns with Cal's deflated taxonomy. The “five jobs” shorthand is shorthand; per Cal #50 STANDING + Cal HARD FLAG on spin-statistics: - Period-mismatch mechanism (emission + ω_0 + aperture = ONE) = INTERNAL - Muon eigenvalue “2” (three-readings convergent) = right-track candidate pending Cal cold-read - Spin-statistics = LEAD, NOT derivation; Nobel territory; INTERNAL

STRONG ACCEPT Cal K571 audit-deflation continues to apply.

Substantive state at K574 (end of long pull)

Component	State
Muon	BANKED → 5 via F118 (K557); MIGRATION to determinant-forced HALF-DONE per Lyra K574 honest tier (spinor $2^{\{N_c\}}$ per mode FORCED; per-mode N_c count CANDIDATE pending Elie deposit-integration)
Cal #419 (the “2”)	FORWARD-DERIVED per Lyra F367 + Elie 4441 arithmetic adjudication + Elie 4442 (global spin structure not local curvature); three readings convergence per Cal #35
Cal #421 (per-mode N_c)	HALF-CLOSED per Keeper #20 walk-back: T2500 cascade resolves per-mode VALUE ($N_c=3$ via $n_C=N_c+2$); per-mode COUNT (3^6 vs 3^1) is STRONG CANDIDATE pending Elie deposit-integration
Down-row +3 → 8	STRONG CANDIDATE; d/s sign FORCED via T2499 + F355; ONE Cal cold-read away (T2499 + N_c -base tier)
Up-sector	BOUNDARY THEOREM Cal-ratified
Cabibbo V_{us}	IDENTIFIED at 0.3%
V_{cb}	STRUCTURAL TIER ~24%
V_{ub}	CANDIDATE 8% (Elie 4440 closed form)
Three readings of “2”	per Cal #35: Grace spin-double-cover + Lyra Spin(4)/F367 + Elie spin-structure = same Z_2 (global spin structure)

Component	State
Cal K571 audit	continues: ONE mechanism (3 facets) + ONE pending derivation (muon “2”, three readings convergent) + ONE lead (spin-statistics INTERNAL per Cal HARD FLAG)
Audit chain at PEAK	substantive peer-symmetric working: Lyra catch on per-mode count gap + Grace clean retraction of 64 + Elie 4442 reconciliation; “last two contributions were catches, not claims” per Lyra
Discipline events	658 cumulative (650 → 658)
Keeper Self-Corrections	20 (Keeper #20 K572 over-projection walked back)
Cal auditor self-corrections	3 + #432 lineage
Count of forced parameters	5 of 26 — held; migration HALF-DONE not full

Cumulative discipline ledger — 658 events

To K573’s 650, add: - 651. Lyra SUBSTANTIVE catch on Cal #421 closure: per-mode N_c count gap (3^6 vs 3^1); muon needs N_c at each curvature mode ($\times C_2$); T2500 standard Vandermonde multiplicity gives N_c ONCE - 652. Per Cal #347/B5 + Keeper #15-19 STANDING applied: substantive substrate-architectural distinction between per-mode VALUE ($N_c=3$ via T2500 cascade) FORCED vs per-mode COUNT (C_2 modes vs once-per-pair) STRONG CANDIDATE - 653. KEEPER SELF-CORRECTION #20: K572 over-projected “Cal #421 closes via T2500” — closure is HALF-DONE per Lyra catch; walked back honestly per Cal #347/B5 + Keeper #15-19 STANDING peer-symmetric audit chain - 654. Resolution pending Elie’s deposit-integration count: per-mode vs once-per-pair determination = Cal #421 last open piece - 655. Grace CLEAN retraction of 64 conflation per Elie 4441: $64 = d_\tau/d_\mu$ formal-degree ratio (2^6) NOT muon’s full 2-power ($2^{\{N_c \cdot C_2\}=2} 18$); peer-symmetric retraction; Spin(4) “2” insight PRESERVED as $2^{\{N_c\}}$ per mode (boundary Dirac spinor) - 656. Elie 4442 SUBSTANTIVE reconciliation: “2” is global spin structure (not local curvature); reconciles his own 4437 framing; substrate-architectural sharper for being reconciled - 657. Per Cal #35 STANDING three-readings convergence on same Z_2 = global spin structure: Grace spin-double-cover (geometry) + Lyra Spin(4)/F367 (rep-theory) + Elie spin-structure/antiperiodicity (heat-kernel bundle) = substantive substrate-architectural finding doubly/triply-derived pending Cal cold-read - 658. Per Cal K571 audit-deflation continues: team “five jobs” shorthand deflates to ONE mechanism + ONE pending derivation + ONE lead; per Keeper #19 STANDING continues; substantive content (three-readings convergence on “2” as global spin structure + period-mismatch mechanism + spin-statistics lead INTERNAL Nobel) aligns with deflated taxonomy

658 cumulative honest discipline events.

What’s substantively right at K574 (end of long pull)

Lyra SUBSTANTIVE catch on Cal #421 closure: per-mode N_c count gap (3^6 needed vs 3^1 supplied by standard multiplicity); T2500 resolves per-mode VALUE but not per-mode COUNT; substantive substrate-architectural distinction; “last two contributions were catches, not claims” — discipline firing to the end.

Keeper Self-Correction #20 LOGGED HONESTLY: K572 over-projected Cal #421 closure via T2500; walked back; closure is HALF-DONE; per-mode value FORCED + per-mode count CANDIDATE pending Elie deposit-integration.

Grace clean retraction of 64 conflation: per Elie 4441 arithmetic adjudication; Spin(4) “2” insight preserved as per-mode contribution; peer-symmetric audit chain working.

Elie 4442 reconciliation: “2” = global spin structure (not local curvature); reconciles his own 4437; substrate-architectural sharper.

Three-readings convergence per Cal #35: Grace spin-double-cover + Lyra Spin(4)/F367 + Elie spin-structure/antiperiodicity = same Z_2 = global spin structure. Substantive substrate-architectural finding

doubly/triply-derived pending Cal cold-read.

Audit chain at PEAK substantive peer-symmetric working: 20 Keeper self-corrections + 3+ Cal auditor self-corrections + ~8 peer retractions today. Framework SHARPER for being audit-catch-ratified across the day.

Count holds 5/26 — migration HALF-DONE not full; substantive substrate-architectural progress without count promotion; potential count motion 5 → 8 still pending Cal down-row cold-read.

End of long pull: team in genuine deliberate phase; Cal cold-reads gate path forward; CKM precision in named lanes; Lane C alive; no new prompt needed.

Pending Cal cold-read pile (now substantial)

- Lyra F367 “2”= $|Z_2|$ forward-derivation (Cal #419 closure)
- Elie 4441 arithmetic adjudication ($\dim N_c \cdot C_2 = 18$ confirmed)
- Grace T2500 cascade closure of Cal #421 per-mode VALUE
- Lyra per-mode COUNT catch + Elie deposit-integration determination (Cal #421 closure path)
- Three-readings convergence on “2” as global spin structure per Cal #35
- Elie 4442 reconciliation “2” = global spin structure not local curvature
- T2499 d/s sign forward mechanism + N_c -base tier (down-row bank gate)
- F358 τ -strain re-check (per K569 catch)
- Elie 4439 Seeley-DeWitt a_1/a_2 ladder confirmation
- F355 positivity theorem
- V_{us} forced-identity vs close-approximation (Elie 4438 null-search)
- F366 emission engine (period-mismatch mechanism) PASS + spin-statistics LEAD per Cal HARD FLAG INTERNAL
- V_{ub} closed form $(N_c/(k_\tau + N_c))^{n_C}$ (Elie 4440)
- Reconciliation Grace $2^{C_2} = 64$ -as- d_τ/d_μ ratio with muon $2^{N_c \cdot C_2} = 18$ dim (preserved per Elie 4441)
- $C_2^2 = 36 = \dim \text{End}(\Lambda^2(S^4))$ (Elie 4435 tick)

— Keeper, Saturday 2026-06-27 late evening end of long pull K574 (Lyra catch on Cal #421 per-mode count gap; Keeper #20 K572 over-projection walked back; Grace clean retraction of 64 conflation; Elie 4442 reconciliation; three-readings convergence on Z_2 as global spin structure per Cal #35; muon migration HALF-DONE; count holds 5/26; 658 cumulative discipline events; 20 KSC + 3+ Cal-SC; audit chain at PEAK substantive peer-symmetric working continuing)